FINITE PROJECTIVE MODULES OVER THE FIBRE PRODUCT OF RINGS CORRESPOND TO MATCHING PAIRS OF FINITE PROJECTIVE MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A module is finite projective if and only if it is finitely presented and flat. Algebra, Lemma \ref{algebra-lemma-finite-projective}, cited at proof lines 903-904. Reduces the equivalence to matching finite presentation and flatness.
- Flatness and finiteness of the module associated to a triple over the fibre product. Lemmas \ref{lemma-flat-module-over-fibre-product} and \ref{lemma-finite-module-over-fibre-product}, cited at proof lines 905-907 and again at 914 and 929. Reduce to finite presentability of the fibre product module and provide the finiteness input for the kernel.
- The kernel of a surjection from a finite free module onto a finite module is finite. Algebra, Lemma \ref{algebra-lemma-extension}, cited at proof lines 927-928. Shows that the kernels after base change are finite modules.
- Flatness makes the tensor product of a short exact sequence with a module exact. Algebra, Lemma \ref{algebra-lemma-flat-tor-zero}, cited at proof lines 936-938. Gives the identifications K_B tensor A = K_A = K_{A'} tensor A that the finiteness criterion needs.

Source reference. Original lemma 0D2J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16384–16384. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10885–10885. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D2I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16383–16383. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D2H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16382–16382. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0519: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6527–6527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 647–647. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A, A', B, B', I be as in Situation 08KH. The category of finite projective B'-modules is equivalent to the full subcategory of $Mod_B \times_{Mod_A} Mod_{A'}$ consisting of triples (N, M', φ) with N finite projective over B and M' finite projective over A'.*

Proof. Recall that a module is finite projective if and only if it is finitely presented and flat, see Algebra, Lemma 00NX. Using Lemmas 0D2I and 0D2H we reduce to showing that $N' = N \times_{\varphi,M} M'$ is a B'-module of finite presentation if N finite projective over B and M' finite projective over A'.

By Lemma 0D2H the module N' is finite over B'. Choose a surjection $(B')^{\oplus n} \to N'$ with kernel K'. By base change we obtain maps $B^{\oplus n} \to N$, $(A')^{\oplus n} \to M'$, and $A^{\oplus n} \to M$ with kernels K_B, $K_{A'}$, and K_A. There is a canonical map

$$K' \longrightarrow K_B \times_{K_A} K_{A'}$$

On the other hand, since $N' = N \times_{\varphi,M} M'$ and $B' = B \times_A A'$ there is also a canonical map $K_B \times_{K_A} K_{A'} \to K'$ inverse to the displayed arrow. Hence the displayed map is an isomorphism. By Algebra, Lemma 0519 the modules K_B and $K_{A'}$ are finite. We conclude from Lemma 0D2H that K' is a finite B'-module provided that $K_B \to K_A$ and $K_{A'} \to K_A$ induce isomorphisms $K_B \otimes_B A = K_A = K_{A'} \otimes_{A'} A$. This is true because the flatness assumptions implies the sequences

$$0 \to K_B \to B^{\oplus n} \to N \to 0 \quad \text{and} \quad 0 \to K_{A'} \to (A')^{\oplus n} \to M' \to 0$$

stay exact upon tensoring, see Algebra, Lemma 00HL. □